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/Users/Keys/.local/share/virtualenvs/prob-and-stat-project-2-8opPzMJz/bin/python /Users/Keys/Projects/prob-and-stat-project-2/analysis.py
SAMPLE SIZES
Marsh: 199
Galton: 205

SAMPLE MEANS
Marsh Mean male height: 1732.4925 mm
Marsh Mean female height: 1601.9497 mm
Galton Mean male height: 1760.6289 mm
Galton Mean female height: 1625.6496 mm

SAMPLE STANDARD DEVIATIONS (StDev.S)
Marsh male height standard deviation: 68.7507 mm
Marsh female height standard deviation: 62.4350 mm
Galton male height standard deviation: 67.2387 mm
Galton female height standard deviation: 59.2590 mm

SAMPLE MEAN CORRELATION
Correlation between male and female heights (Marsh): 0.3644
Correlation between male and female heights (Galton): 0.0969

THEORETICAL DIFFERENCE (Bivariate Normal Model)
Marsh E(x1-x2) = 130.5427 mm
Marsh Var(x1-x2) = 5496.1585 mm^2
    std of that is 74.1361 mm
(Marsh) Z score of difference=0: -1.7609
P(z > -1.7609): 0.9609
Galton E(x1-x2) = 134.9793 mm
Galton Var(x1-x2) = 7260.4819 mm^2
    std of that is 85.2085 mm
(Galton) Z score of difference=0: -1.5841
P(z > -1.5841): 0.9434

SAMPLE DIFFERENCE (Paired Y values)
Marsh height difference mean: 130.5427 mm
Marsh height difference std: 74.1361 mm
(Marsh) Z score of difference=0: -1.7609
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P(z > -1.7609): 0.9609
(Marsh) Actual positive difference proportion: 0.9447
Galton height difference mean: 134.9793 mm
Galton height difference std: 85.2085 mm
(Galton) Z score of difference=0: -1.5841
P(z > -1.5841): 0.9434
(Galton) Actual positive difference proportion: 0.9171

HYPOTHESIS TESTING (Single Dataset)
Marsh, Test Statistic (z): 0.0000
    Test statistic is within 95% confidence interval (Fail to reject H0).
    Test statistic does not suggest proportion is greater than model.
Galton, Test Statistic (z): -0.0000
    Test statistic is within 95% confidence interval (Fail to reject H0).
    Test statistic does not suggest proportion is greater than model.

JOINT HYPOTHESIS TESTING (Marsh vs Galton)
Joint test statistic (z): 0.8205
    Statistic 0.8205 is within (-1.9600, 1.9600).
    No significant cultural difference detected.

T TESTING (Difference in average height by sex across datasets is significant)
Male test statistic (t): -4.1573. P-value: 0.0000
    Significant difference detected between datasets.
Female test statistic (t): -3.9111. P-value: 0.0001
    Significant difference detected between datasets.

T TESTING (Difference in heights pairwise is significant)
Marsh Dataset test statistic (t): 24.8399. P-value: 0.0000
    Significant difference detected between heights
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Galton Dataset test statistic (t): 22.6810. P-value: 0.0000
    Significant difference detected between heights

Process finished with exit code 0

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